An Aerosol Retrieval Algorithm Over Land From Multispectral Single-Viewing Measurements of Intensity and Polarization

Zhe Ji, Zhengqiang Li, Zihan Zhang, Gerrit de Leeuw, Yan Ma, and Zheng Shi

***Abstract*— Currently, there is no generalized optimization algorithm specifically designed for single-viewing-angle polarimeters. To better explore the potential retrieval capabilities of such sensors, we have developed a generalized optimized aerosol retrieval framework. To further validate its effectiveness, we applied this framework to Chinese particulate observing scanning polarimeter (POSP), launched in September 2021. POSP is the first high-precision spaceborne cross-track scanning polarimeter, featuring a broader spectral range (UV-VIS-NIR-SWIR) and full-spectrum polarization capabilities. POSP measurements between January 2022 and December 2022 have been used to generate an aerosol product. Validation with ground-based AErosol RObotic NETwork (AERONET) aerosol products shows good consistency, for AOD with an root mean square error (RMSE) of 0.086, bias of 0.003, and the fraction within the Global Climate Observing System (GCOS) requirements is 58.73%; for AE with an RMSE of 0.404, and the fraction within the expected error (EE) is 71.61%; for SSA with RMSE of 0.067, and the fraction within the EE is 76.06%. Additionally, we compared POSP and MODIS/Terra AOD products. The spatial distributions of POSP and MODIS-DB AOD were similar, exhibiting good consistency in highly polluted areas, while the POSP AOD revealed more detailed information. The information content analysis shows that adding polarimetric measurements can effectively improve the degrees of freedom for signal (DFSs) available for aerosol retrieval, especially for coarse-mode dominated aerosols. This is confirmed by the actual retrieval results showing that adding polarimetric measurements effectively reduces underestimation. Error analysis shows that adding polarization results in a reduction in the uncertainty of POSP AOD at AOD < 0.2 and over sparsely vegetated areas (NDVI < 0.1). The validation and analysis results both demonstrate that the retrieval framework used in this study effectively utilizes both intensity and polarization, and that adding polarization significantly enhances the retrieval accuracy.**

***Index Terms*— Aerosol optical depth, aerosol retrieval algorithm, particulate observing scanning polarimeter (POSP), polarization, satellite remote sensing.**

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I. Introduction

ATMOSPHERIC aerosols are primarily composed of particles and droplets suspended in the atmosphere [1]. Their radius and chemical composition evolve through processes like condensation and coagulation [2]. Aerosols are generally emitted directly into the atmosphere through natural or anthropogenic processes such as sea spray aerosol, or desert dust aerosol, with different source strengths. Additionally, aerosols can be indirectly formed from precursor gases such as NO_2, SO_2, NH_3, VOCs, which also are emitted from a large variety of sources. This results in strong spatiotemporal variation of the concentrations and chemical composition of aerosol particles. Furthermore, these particles have relatively short atmospheric lifetimes (hours to days). Aerosol particles with specific chemical compositions can exhibit hygroscopic properties and serve as strong cloud condensation nuclei (CCN), readily activating into cloud droplets under low supersaturation conditions. These particles, often containing water-soluble compounds such as sulfates, play a crucial role in influencing both the micro- and macrophysical characteristics of clouds [3], [4]. As a result of the above, aerosols remain one of the largest uncertainties in the Earth' climate system, as reported by intergovernmental panel on climate change (IPCC) assessments [5], [6], [7]. Furthermore, aerosols also are harmful to human health. Inhalation by human beings of fine particles produced from events like forest fires or industrial activities can lead to severe lung infections [8], [9].

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Aerosol parameters can be obtained through ground-based or satellite measurements. Ground-based measurements provide information with high accuracy but limited to a certain region. In contrast, satellite measurements cover a large area but generally have lower accuracy than ground-based measurements. Ground-based observations primarily rely on sun/sky radiometers, which measure direct and diffuse sunlight. Established ground-based networks, such as NASA's AErosol RObotic NETwork (AERONET) [10], Europe's PHOtométrie pour le Traitement Opérationnel de Normalization Satellitaire (PHOTONS) [11], and China's Sun–Sky Radiometer Observation NETwork (SONET), which provide unique polarimetric observations [12].

The use of satellite measurements for aerosol retrieval began in the 1970s [13]. Aerosol retrieval methods can be broadly categorized into two types: based on single or multiviewing angles observations. Early studies primarily focused on intensity from single-view sensors, but numerous studies have shown that the use of multiviewing angles [14], [15] and polarization are equally important for improving aerosol retrieval accuracy [16], [17] (Mishchenko and Travis found that utilizing polarization enables effective aerosol retrieval, particularly for the microphysical characteristics of aerosols [18]). The advanced very high resolution radiometer (AVHRR) [19], the dual-view of the along-track scanning radiometer (ATSR-2 on ERS-2) [20], and the advanced ATSR (AATSR on ENVISAT) [21] were among the early sensors used for aerosol retrieval, primarily relying on look-up table methods. Subsequent advancements came with the Moderate Resolution Imaging Spectroradiometer (MODIS), which provides 32 spectral bands between 0.405 and 14.385 μm. Kaufman et al. [22] introduced the dark target (DT) algorithm, suitable for primarily vegetated ("dark") land surfaces. Additionally, the deep blue (DB) algorithm utilizes historical data to statistically derive surface reflectance for various bands [23]. The multiangle implementation of atmospheric correction (MAIAC) algorithm considers surface directional characteristics and simultaneously retrieves aerosol parameters along with surface directional properties by using MODIS time series [24], [25]. The multi angle imaging spectroradiometer (MISR), which is on the same satellite platform as MODIS, offers fewer spectral bands but provides more observation angles (than MODIS). Martonchik et al. [14] successfully used this multiangle information to retrieve aerosol parameters. The polarization and directionality of the Earth's reflectance (POLDER) instrument is the world's first spaceborne polarimeter, featuring both polarization and multi-angle observation capabilities. Deuzé et al. [16] was the first to achieve aerosol retrieval based on POLDER measurements. Tanré et al. [17] updated this algorithm, providing higher precision results. Hasekamp et al. [26] developed an aerosol optimization retrieval algorithm over ocean using POLDER data, successfully retrieving aerosol parameters. Dubovik et al. [27] also used POLDER measurements to implement simultaneous retrieval of aerosol and surface parameters based on advanced statistically optimized fitting.

TABLE I
METRICS OF POSP AOD OVER LAND

AOD range	RMSE	MAE	Bias	GOCS(%)
Full	0.0860	0.0540	0.0030	58.73
[0-0.2]	0.0560	0.0375	0.0165	67.26
[0.2-0.7]	0.1404	0.1090	-0.0464	24.05
[0.7-3]	0.2333	0.1872	-0.0919	31.54

China has launched a number of environmental monitoring satellites [28], [29], [30], including polarimeters [31], [32], [33], [34], [35]. The particulate observing scanning polarimeter (POSP) is a novel sensor mounted on the GF-5(02) satellite, which has been launched in September 2021. Aerosol algorithms have already been developed based on POSP intensity measurements [36], [37], [38]. Although Chen et al. [39] have explored the use of the polarimetric measurements offered by POSP, the accuracy of its AOD retrieval still needs improvement. Therefore, further exploring the potential of POSP in aerosol parameter retrieval remains highly valuable.

In our previous research, we developed an aerosol retrieval algorithm for POSP's intensity observations [37], but it did not take into account POSP's unique polarization observations. In the current study, the use of an optimization retrieval technique that integrates intensity and polarization is explored from POSP. For intensity measurements, we continued to use our previous retrieval framework [36], [37] and added measurements at 670 and 865 nm. The same method used in previous work was applied to reconstruct the BRDF kernel coefficients for these two bands and analyze the empirical relationships between various parameters. Validation against AERONET data indicates that POSP retrievals with high accuracy, for AOD with an root mean square error (RMSE) of 0.086 and the fraction within the expected error (EE) of 77.64%; for AE with an RMSE of 0.404, and the fraction within the EE is 71.61%; for SSA with an RMSE of 0.067, and the fraction within the EE is 76.06%. Furthermore, to detect POSP's ability to monitor pollution events, we selected typical highly polluted regions, including the North China plain (NCP) and Indian Peninsula. Compared to the MODIS AOD product, POSP AOD provides more detailed pollution information, demonstrating better pollution monitoring capabilities under these conditions. To further analyze the impact of polarization on retrieval accuracy, we applied the optimal estimation theory and calculated the degree of freedom for signal (DFS) available for aerosol retrieval parameters to quantify their information content [40], [41]. The results indicate that adding polarization significantly increases the information content of aerosol parameters, with a more pronounced improvement for coarse-dominated aerosols compared to fine-dominated aerosols.

This study not only serves as a valuable addition to the existing applications of POSP data but also provides a potential development direction for future aerosol retrieval using spaceborne single-viewing polarimeters.

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II. Materials

A. POSP Data

The POSP is a single-view scanner, and it features a field of view of ±50° with a nadir resolution of 6.4 km and a swath width of approximately 1850 km, with a local equator crossing time of 10:30 a.m. in descending mode. Calibration results indicate radiometric calibration accuracy of better than 5% and a degree of linear polarization (ΔDoLP) precision of better than 0.005 [35]. The POSP can simultaneously detect Stokes vector [I, Q, U] of the nine-band at 380, 410, 443, 490, 670, 865, 1380, 1610, and 2250 nm. Unlike most existing or former onboard polarimeters which shown in Table S1, the POSP additionally provides unique polarimetric measurements in the shortwave infrared (SWIR) alongside the UV/visible spectrum (UV/VIS).

B. Aeronet Data

The AERONET is a ground-based remote sensing aerosol networks, utilizing sun photometers to measure direct and scattered sunlight at eight wavelengths (340, 380, 440, 500, 675, 870, and 1020 nm) at 15-min intervals (https://aeronet.gsfc.nasa.gov/). It offers AOD products with uncertainties of 0.01 in the VIS and 0.02 in the UV [42]. POSP AOD are reported at 550 nm, and for validation, it is essential to interpolate the AERONET AOD to this specific wavelength. For improved accuracy in estimating AOD at 550 nm, quadratic polynomial fitting is considered more reliable than linear interpolation using the Ångström exponent (AE) [43]. This interpolation is carried out using an empirical equation relating AOD to the observed wavelengths

$$\log(\tau_\lambda) = a_0 + a_1 \log(\lambda) + a_2 \log(\lambda) \tag{1}$$

where λ is the specified wavelength in microns, τ_λ is the AOD at wavelength λ, and a_0, a_1, and a_2 are the fitting coefficients.

In this study, Level 1.5 data are selected as the ground-based reference data for AOD and AE, and Level 2.0 data for SSA.

C. MODIS Data

MODIS has been in orbit for over two decades [44], flying on the Terra satellite launched in December 1999, with an equator crossing time at 10:30 a.m. in a descending mode, and on the Aqua satellite launched in May 2002, with an equator crossing time of 13:30 in an ascending mode. The MAIAC algorithm utilizes the synergistic observations of both MODIS/Terra and MODIS/Aqua for retrieval, providing greater spatial coverage compared to the other two algorithms [45]. However, as the POSP algorithm depends solely on observations from a single sensor, we only employ the DB algorithms for comparative analysis in Section IV-B (DT does not include results for bright surface). This study utilizes MODIS Collection 6.1 (C6.1) (https://data.cresda.cn/).

III. Methodology

A. Basic Principle

1) Apparent Reflectance: The reflectance ρ is defined as follows:

$$\rho = \frac{\pi I}{E_0 \mu_s} \tag{2}$$

where E_0 is the flux density at TOA and μ_s represents the cosine of the solar zenith angle (θ_s). The intensity of the reflectance at TOA (ρ^*) is calculated using the non-Lambertian radiative transfer model proposed by Qin et al. [46]. This model is widely used for aerosol retrieval, as it separates the information of the surface and the atmosphere in the integral calculation [47], [48]

$$\rho^* = \rho_0 + \frac{\vec{T}(\mu_s)^T R \vec{T}(\mu_v) - e^{-\tau/\mu_s}|R|e^{-\tau/\mu_v}}{1 - \rho_{\mathrm{BHR}} S} \tag{3}$$

where ρ_0 denotes the atmospheric path reflectance, μ_v represents the cosine of the viewing zenith angle (θ_v), $\vec{T}(\mu_s)^T$ is the downward transmission vector, and $\vec{T}(\mu_v)$ is the upward transmission vector, R represents the different hemispherical reflectance, τ is the atmospheric optical depth, ρ_{BHR} is the bi-hemispherical reflectance (BHR) equal to the surface albedo, and S is the spherical albedo of the atmosphere. The definition of the parameters is same as in [37].

2) Apparent Polarized Reflectance: The apparent polarized reflectance ρ_p is defined as follows:

$$\rho_p = \frac{\pi \sqrt{Q^2 + U^2}}{E_0 \mu_s} \tag{4}$$

where Q and U represent the radiance of linear polarization. The DoLP is DoLP $= (Q^2 + U^2)^{1/2}/I$. To calculate the polarized reflectance at the TOA (ρ_p^*), we utilized the model developed by [49]

$$\rho_p^* = \rho_{p0} + T^{\downarrow}(\mu_s) R_p T^{\uparrow}(\mu_v) \tag{5}$$

where ρ_{p0} denotes the polarized path reflectance, which includes molecular and aerosol contributions, R_p is the surface polarized reflectance, $T^{\downarrow}$ and $T^{\uparrow}$ describe the downward and upward transmission, respectively.

B. Aerosol Model

The ratio between the radius of an aerosol particle and the wavelength of incident radiation determines the scattering properties of that particle [50], assuming a spherical particle with a given complex refractive index ($m = n + ik$). The scattering of incoming radiation is determined by the particle size distribution. In this study, a bi-modal log-normal volume size distribution is used

$$\frac{dV}{d\ln r} = \sum_{i=1}^{2} \frac{V_0^{(i)}}{\sqrt{2\pi}\ln\sigma_g^{(i)}} \exp\left[-\frac{\left(\ln r - \ln r_g^{(i)}\right)^2}{2\ln^2\sigma_g^{(i)}}\right] \tag{6}$$

$$\begin{cases} r_{\mathrm{eff}} = r_g \exp\left(-\dfrac{1}{2}\ln^2\sigma_g\right) \\ v_{\mathrm{eff}} = \exp\left(\ln^2\sigma_g\right) - 1 \end{cases} \tag{7}$$

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where V_0 represents the normalized column volume concentration, r_g and σ_g are the median radius and the standard deviation, respectively. r_{eff} and v_{eff} represent the effective radius and the effective variance, respectively.

C. Surface Model

As in previous studies [37], the Ross-thick Li-sparse (RTLS) model is used to compute the surface reflectance [37]. The RTLS model consists of three main components: isotropic kernel (K_{iso}), volumetric kernel (K_{vol}), and geometric-optical kernel (K_{geo}) [51], [52]

$$R_{\text{BRDF}} = f_{\text{iso}} K_{\text{iso}} + f_{\text{vol}} K_{\text{vol}} + f_{\text{geo}} K_{\text{geo}} \tag{8}$$

where R_{BRDF} is the surface binomial reflectance distribution function, f_{iso}, f_{vol}, and f_{geo} are the corresponding kernel coefficients. The calculation of kernel functions is presented in the (S5)–(S13).

This study also adopts the Maignan BPDF model. Maignan established a new BPDF model based global polarimetric measurements with the POLDER [53]. Due to the Maignan BPDF model requiring fewer fitting parameters and demonstrating higher accuracy [54], it has been widely applied in retrieval algorithms

$$R_{\text{BPDF}} = \frac{\alpha \exp(-\tan\theta_r)\exp(-\text{NDVI})F(\theta_r, m)}{4(\mu_s + \mu_v)} \tag{9}$$

where α is a free parameter, depending on the surface type. $F(\theta_r, m)$ denotes the Fresnel reflection, θ_r denotes one half of the phase angle, m is the reflective index for land surface, and 1.5 is adopted [55]. The calculation of $F(\theta_r, m)$ can be found in (S14)–(S17).

D. Retrieval Algorithm

In earlier research, Ji et al. [37] achieved high-precision aerosol retrieval based on POSP intensity measurements. In the current study, that work is extended. First, a BRDF kernel coefficient library for 670 and 865 nm has been constructed using the MODIS BRDF product (MCD43). Therefore, we added more intensity bands (670 and 865 nm) to the current algorithm. Second, we achieve the joint inversion of intensity and polarization by adding polarization observation constraints to the optimization iteration process. Finally, aerosol parameters and surface parameters can be synchronized retrieved by updating the global aerosol model distribution (The description of the updated global aerosol model can be found in [37]). We did not use shorter wavelengths (380 and 410 nm) to simplify the computation of the LUT, and reduce the error caused by a fixed aerosol layer height (ALH). Meanwhile, the Rayleigh scattering dominates at these wavelengths using a fixed ALH will lead to large errors (the error induced by the fixed ALH tends to decrease with increasing wavelength [23], [56]). By reading the HDF5 files of POSP data, the measurements and observation geometries required for retrieval can be extracted pixel by pixel. After extracting the measurement, the specific retrieval process is as follows (see flowchart in Fig. 1).

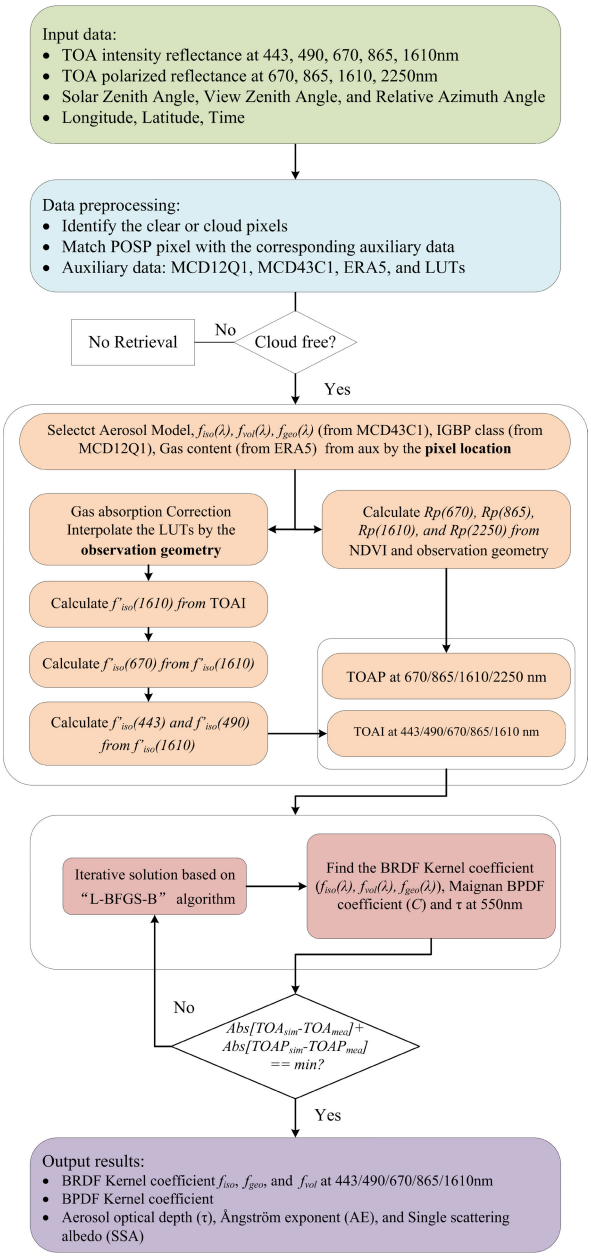


Fig. 1. Flowchart of the main steps of the POSP aerosol retrieval algorithm.

1) *Data Preprocessing:* correction for gaseous absorption using ERA5 reanalysis data selected for the location of the pixel and observation geometries (the details for preprocessing are presented in supplementary).
2) *Calculation of Surface Reflectance:* the surface reflectance are calculated using (8)–(9). The optimization algorithm requires setting prior values for process beginning. Specifically, the BRDF kernel coefficients at 443 and 490 nm are propagated from 1610 to 670 nm and then to 443 and 490 nm, while the other parameters are initialized using historical statistical values.
3) *Linear Interpolation LUT:* based on the observation geometry and altitude, the atmospheric parameters [in (2)–(6)] corresponding to different aerosol models

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are extracted. Then, following the method proposed by Ji et al. [37], the reflectance at the TOA is expressed as a linear function of AOD.

4) *Constructing the Cost Function:* the reflectance at the TOA can be calculated using (2) and (6). Then we get the cost value from (10), and we linearize the interpolated results to obtain the reflectances at TOA that varies solely with AOD.
5) *Iterative Inversion:* The state vector, which includes AOD, the isotropic kernels at 443, 490, 670, 865, and 1610 nm, and BPDF parameter was iteratively optimized. The state vector that minimizes the cost function is then taken as the final retrieval result for the pixel.

In order to synchronize the intensity and polarimetric measurements in the algorithm, it is necessary to extend the existing cost function [37, eq. (6)] (The previous cost function only considered intensity observations, ignoring the differences in weights between intensity and polarization), as shown in the following equation:

$$\text{Cost} = \sqrt{\sum_{i=1}^{5} \frac{\left(\rho^*_{\lambda_i} - \widehat{\rho^*_{\lambda_i}}\right)^2}{S_I(i)}} + \sqrt{\sum_{k=1}^{4} \frac{\left(\rho^*_{p_{\lambda_k}} - \widehat{\rho^*_{p_{\lambda_k}}}\right)^2}{S_P(k)}} \quad (10)$$

where $\widehat{\rho^*_{\lambda_i}}$ and $\rho^*_{\lambda_i}$ represent the simulated [by (3)] and observed reflectance at TOA, respectively. $\widehat{\rho^*_{p_{\lambda_k}}}$ and $\rho^*_{p_{\lambda_k}}$ represent the simulated [by (5)] and observed polarized reflectance at TOA, respectively. $S_I(i)$ and $S_P(k)$ denote the intensity and polarization measurement error, respectively.

We use the optimization algorithm "L-BFGS-B" from Python library SciPy to solve the cost function [57], [58]. In addition, the statistical errors are also considered by constructing boundary constraints during the optimization iteration process as follows.

1) *Instrument Errors:* The instrument error indicates the range of the permissible error between simulations and measurements. Therefore, we use calibration accuracy as the error margin: for intensity measurements, we set $\text{abs}(\rho^*_{\lambda_i} - \widehat{\rho^*_{\lambda_i}})/\rho^*_{\lambda_i} < 0.05$; for polarimetric measurements, we set $\text{abs}(\rho^*_{p_{\lambda_k}} - \widehat{\rho^*_{p_{\lambda_k}}})/\rho^*_{\lambda_i} < 0.005$.
2) *Model Errors:* The empirical relationships (the relationships of BRDF kernel coefficient between 443/670 nm and 490/865 nm) which have been used for surface constraints that contain some errors. These errors indicate the range of permissible bias between simulations and observations. The relationship between the BRDF kernel coefficients should satisfy the precomputed range: for 443 and 490 nm, $F(f'_l(443), f'_l(490)) < \text{RMSE}(l = \text{iso, val, geo})$; for 670 and 865 nm, $F(f'_l(670), f'_l(865)) < \text{RMSE}(l = \text{iso, val, geo})$.
3) *Empirical Knowledge* The retrieved AOD and surface parameters have seasonal characteristics [21], [53], [54], [59]. Therefore, the above parameters to be inverted should fall within a certain statistical range. For AOD, the MCD19 (from 2019 to 2022), which is the MODIS AOD data product by MAIAC algorithm has been used to determine the possible AOD ranges. For BRDF kernel coefficients, MCD43 (from 2011 to 2022), which provides black-sky albedo and white-sky albedo for MODIS bands 1 through 7 has been used; However, due to the lack of long-term observations of the surface polarized reflectance, we do not constrain the BPDF parameter.

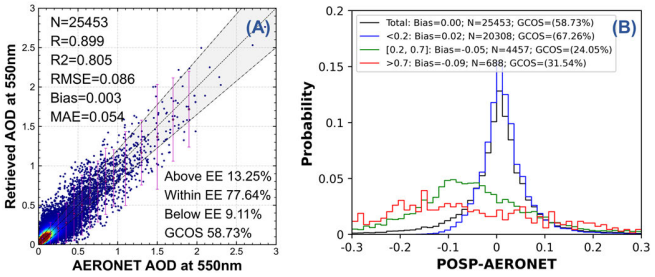


Fig. 2. (a) Density scatter plots of POSP AOD versus AERONET AOD reference data. The black dotted and dashed lines are the 1:1 reference line and the EE lines. The magenta points are means for specific ranges of AERONET and satellite AOD, and the magenta lines are the mean $\pm 2\sigma$ of retrievals in a certain AOD range (binned in intervals of 0.1 for AOD < 1, in intervals of 0.2 for larger AOD). (b) Probability density plot of differences (POSP-AERONET), The black, blue, green, and red solid lines indicate different AOD conditions: Total AOD, AOD < 0.2, $0.2 \leq$ AOD ≤ 0.7, and AOD > 0.7, respectively.

IV. Results

A. Validation of POSP Aerosol Parameters

1) AOD Validation Using AERONET: Fig. 2 shows the validation of POSP AOD retrieval results from January to December 2022, with AERONET data as reference. To ensure consistency with previous retrievals, we followed the same collocation criteria as in Chen et al. [39], where we selected the median AOD from AERONET (with at least two records) within one hour of the GF-5(02) satellite overpass, and the median POSP AOD from a 3×3 window (with a minimum of three retrievals) centered on the AERONET site. To quantitatively evaluate retrieval accuracy, we used several statistical metrics, including the Pearson's correlation coefficient (R), R-squared (R^2), RMSE, mean absolute error (MAE), and Bias [see (S18)–(S22)]. The EE in (S23) is defined as the envelope containing at least 67% (approximately one standard deviation) of the matchups [60]. To satisfy the accuracy requirements for atmospheric research related to remote sensing products, we also applied rigorous criteria set by the Global Climate Observing System (GCOS) in (S25). For AOD, the goal for GCOS is set the greater of 0.04 or 10%, which were also adopted in the Aerosol_cci study [61].

Fig. 2(a) shows a scatter density plot; Fig. 2(b) shows the probability distribution of AOD bias. Overall, the validation shows the strong correlation between POSP AOD and AERONET AOD, with R of 0.899, RMSE of 0.086, Bias of 0.003, fraction within the EE of 77.64% and fraction within the GCOS requirements of 58.73%. Similar to the results obtained in the past based on intensity observations only [37], the probability distribution in Fig. 2(b) indicates that the AOD bias tends to be more uniformly distributed at higher AOD. Specifically, when AOD < 0.2, the probability distribution closely resembles a normal distribution. However, as AOD increases, this characteristic distribution pattern fades.

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Additionally, it is noteworthy that for AOD < 0.2, the fraction within the GCOS requirements (67.26%) is significantly higher than for other AOD ranges.

Table I presents the accuracy metrics of POSP AOD for different AOD ranges. Compared with the results of GRASP and RemoTAP provided by [62], the accuracy of the POSP AOD retrieved in this study is higher for the full AOD range (see Table S4). The higher accuracy is primarily due to the superior accuracy metrics of POSP AOD when AOD < 0.2. However, as Table I shows, for AOD > 0.2, the accuracy significantly decreases, which is mainly due to the use of a fixed aerosol model, which amplifies assumption errors under high AOD. The impact of discrepancies between actual and assumed aerosol models on retrieval errors increases with AOD. Li et al. [35] and Hou et al. [63] through experiments, found that with larger aerosol model errors resulting in less information content of retrieved aerosol parameters. Therefore, as aerosol loading increases, the aerosol model error also increases, leading to a corresponding rise in retrieval errors. In Table S5, we compare the retrieval accuracy before and after adding polarization. The results show that adding polarization is effective in improving retrieval accuracy, with the RMSE decreasing from 0.096 to 0.088 and R^2 increasing from 0.803 to 0.814. Notably, after incorporating polarimetric observations, nearly 2000 additional colocations successfully converged in the retrieval process.

2) AOD Comparisons Between POSP and MODIS Products: Fig. 4 shows the global half-year averaged AOD from January 2022 to June 2022 retrieved by POSP (a) and MODIS (b) and (c), respectively. Globally, anthropogenic aerosols are mainly concentrated in the NCP and northern India. In the NCP, industrial emissions, traffic, and household heating dominate [64] while it is also influenced by desert dust in spring [21]. In northern India, fossil fuel use and biomass burning are the primary sources [65]. Additionally, the coastal regions of Central and Western Africa also exhibit high AOD due to biomass burning [66].

POSP AOD is higher than MODIS AOD in northern Asia and northern North America, and we attribute this difference to the lower spatial resolution of POSP. The spatial resolution of MODIS DB/DT AOD is 10 km, but the spatial resolution of the MODIS measurement can be up to 250 m. Therefore, MODIS has more opportunities to reduce the effect of clouds within 10 km, but the spatial resolution of the POSP AOD is the same as the actual observation resolution of POSP (6.4 km). Hence, if cloud contamination occurs in a POSP pixel the AOD is overestimated. This also explains the lack of results from the POSP AOD in northern South America and Central Africa, where the regions mentioned above are covered by clouds for a long period of time and the lower spatial resolution makes it difficult to recognize cloud-free regions.

It should be noted that compared to DB and DT, MAIAC shows significant underestimation across all regions. MAIAC AOD is noticeably lower than POSP AOD in North Africa, the Arabian Peninsula, and eastern China. Despite these differences, all results consistently indicate significantly high AOD

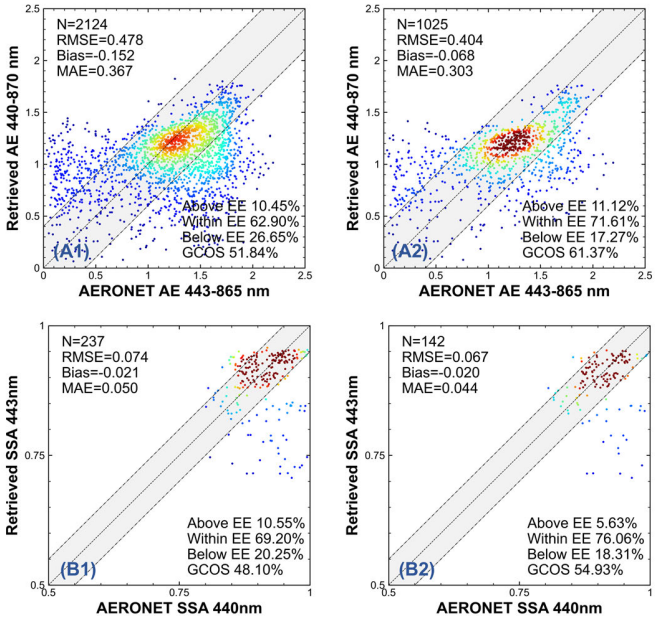


Fig. 3. Validation of POSP retrievals using AERONET as reference. (a1) and (a2) For AE between 443 nm and 865 nm. (b1) and (b2) for SSA at 550 nm. The left column for results when $AOD_{550} > 0.2$ and the right column for results when $AOD_{550} > 0.4$. The black dotted and dashed lines are the 1:1 reference line and the EE lines.

over the Gulf of Guinea in Africa, northern India, and eastern China. Similar discrepancies between satellite products can also be observed in other global comparisons [45], [59]. Due to the lack of AERONET observations in many regions globally, such as in western China and Russia, it is challenging to determine a "global standard" product. Currently, POSP AOD falls between MAIAC and DB, resembling DB over the Gulf of Guinea and northern India, but aligning more with MAIAC over North Africa and the Arabian Peninsula. The comparisons among various satellite products will be further explored in future work.

3) AE and SSA Validation Using AERONET: To further validate the capability of POSP in retrieving aerosol parameters, we also evaluated the retrieved $AE_{443\text{-}865}$ and SSA_{443}. Fig. 3 presents the retrieval accuracy under different AOD conditions. When AOD is low, the aerosol information contained in satellite observations is limited. Therefore, we separately selected the results with $AOD_{550} > 0.2$ or $AOD_{550} > 0.4$ for validation. The validations indicate a significant improvement in retrieval accuracy as the AOD_{550} constraint increases.

For $AE_{443\text{-}865}$, the retrievals are mainly concentrated between 1 and 1.5 (small particle), with a slight underestimation. In contrast, for $AE_{443\text{-}865} < 1$ (large particles), there is a tendency for overestimation. For $AOD_{550} > 0.2$ and $AOD_{550} > 0.4$, the fraction within GCOS requirement (+/−0.3) is 51.84% and 61.37%, respectively. For SSA_{443}, the GCOS requirement (+/−0.03) states that the retrieval error should be smaller than 0.03 [61]. The POSP SSA_{443} retrievals show good consistency with AERONET, with an RMSE of 0.067 and a bias of only −0.02. The fraction within EE requirement is 76.06% when $AOD_{550} > 0.4$.

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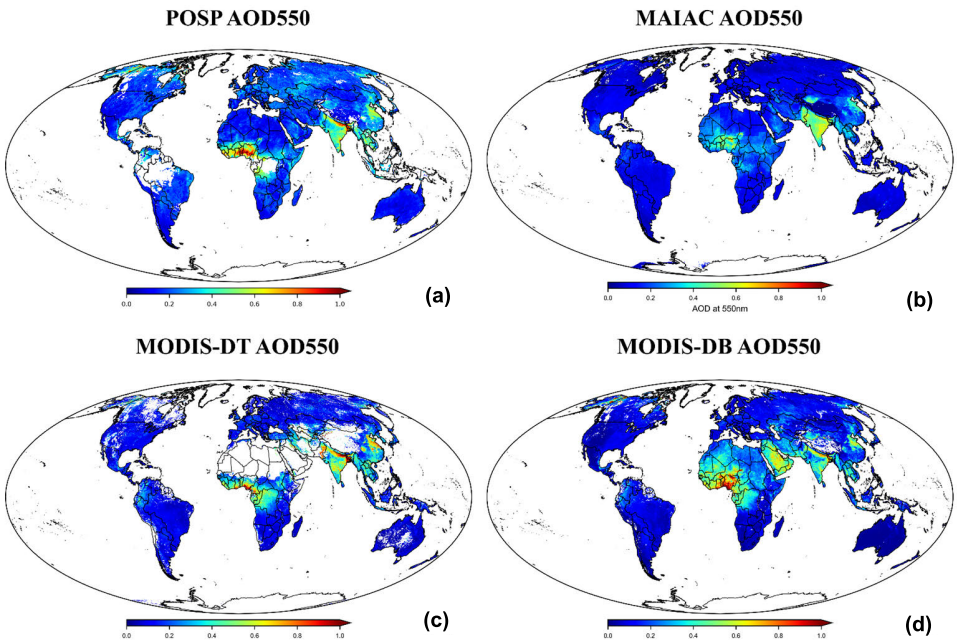


Fig. 4. Spatial distribution of the averaged AOD from January 2022 to June 2022 using (a) POSP, (b) MODIS MAIAC, (c) MODIS DT, and (d) MODIS DB.

B. Global Distribution of POSP Surface Parameters

In this study, both aerosol and surface parameters were retrieved simultaneously. Since the algorithm requires prior values as input during the initial iteration to accelerate retrieval, certain regions lacking historical data were not included in the retrieval process. We will address this limitation in future work. Fig. S5 presents the global surface parameter retrieval results (isotropic kernels at 443, 490, 670, 865, and 1610 nm and BPDF parameter). Given that this study primarily focuses on aerosol parameter retrieval, the validation of surface parameters will be further elaborated in future publications.

C. Results of Case Study

In Fig. 5, POSP false-color images (490 to 670–865 nm), POSP AOD, and MODIS DB AOD over the NCP are presented for September 29, 2022, November 7, 2022, and November 8, 2022. The above period falls in the autumn in the Northern Hemisphere. Potential straw burning, industrial emissions, and regional transportation contribute to significant pollution in China and severe pollution events were observed [64]. POSP and MODIS captured high AOD during two consecutive days as shown in Fig. 5(b) for November 7, 2022, and in Fig. 5(c) for November 8, 2022. It is important to note that, given the errors introduced by the plane-parallel atmosphere assumption at large viewing angles [67], we limited the retrieval with a viewing zenith angle of less than 60° and a sun zenith angle of less than 75°. As a result, in Fig. 5, differences in spatial coverage are observed between the false-color images and POSP AOD.

Compared to November 7, 2022, on November 8, 2022, the AOD extends over a larger area both to the north where it had moved from southern Hebei Province into Beijing and Tianjin, and to the south where it moved from Henan Province into Hubei and Hunan Provinces. The high AOD area was primarily concentrated in Hebei, Henan, and Shandong provinces, forming a continuous polluted area.

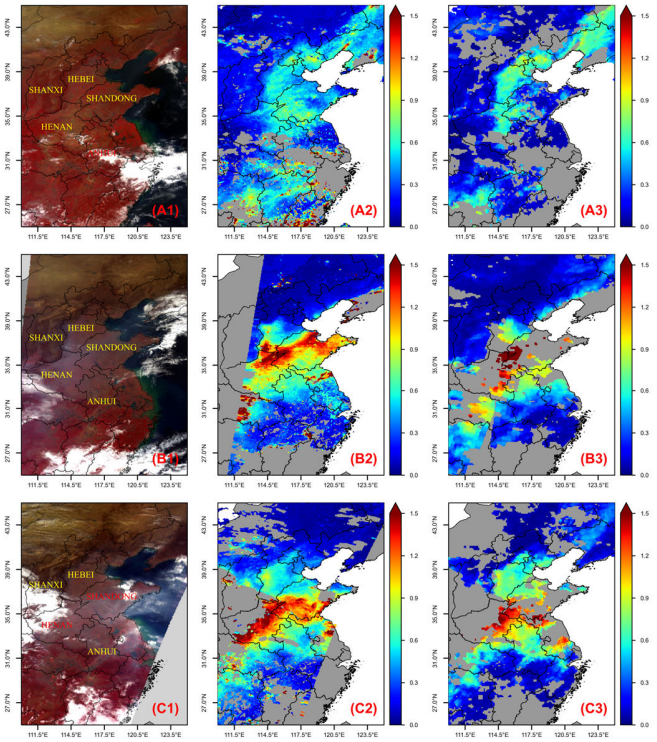


Fig. 5. POSP false color (left column 1) and AOD (at 550 nm) (middle column 2) maps over part of northeast China. For comparison, MODIS DB AOD (at 550 nm) maps are shown in the right column (3). AOD color scale to the rights of columns 2 and 3. The maps show data for (a1)–(a3) September 29, 2022. (b1)–(b3) November 07, 2022. (c1)–(c3) November 08, 2022. Gray color represents land without retrievals.

In addition, compared to POSP AOD, MODIS AOD lacks retrieval results in some regions because POSP identifies these

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regions as having high AOD, whereas MODIS classifies them as clouds. Table S9 shows that on September 29, 2022, and November 8, 2022, the AOD over Beijing was high, while on November 7, 2022, the AOD over Beijing was low. This aligns with the results shown in Fig. 5(a2), (b2), and (c2), where the high AOD over the Beijing area can be clearly observed in Fig. 5(a2) and (c2). Due to the lack of ground-based measurements in southern NCP, we cannot confirm whether the aerosol loading in those areas is similarly high. However, AOD changes continuously and gradually, and MODIS has not classified all high-AOD regions identified by POSP as clouds. MODIS retains some high-AOD regions that overlap with the high-AOD regions identified by POSP. Therefore, this study leans toward the conclusion that a pollution event occurred in the NCP during this period, although the possibility of clouds in this area cannot be ruled out.

In Fig. 5(b1), high AOD areas are evident in Hebei, Henan, and Shandong provinces. Similarly, Fig. 5(b3) shows high AOD in these central areas according to MODIS DB, while POSP AOD captures the full extent of this pollution event. Likewise, Fig. 5(c1) displays a similar situation, though the aerosol plume has spread southwestward. This comparison indicates that MODIS DB AOD maybe mistakenly classifies some high AOD areas as clouds. Therefore, compared to MODIS DB AOD, the POSP AOD shows more details of pollution. For instance, on November 7, 2022, the POSP AOD clearly shows continuous aerosol plumes extending from the border of Hebei, Henan, and Shandong provinces to the southern part of the Bohai Bay. A similar pattern was observed on November 8, 2022, where MODIS DB AOD again failed to provide continuous retrievals for polluted areas. The comparison in Table S9 shown that comparing observations from AERONET sites during this period, the POSP AOD exhibited better retrieval accuracy than MODIS DB on November 7 and 8, 2022. Beijing-PKU showed superior accuracy on September 29, 2022, while other sites experienced underestimations. Therefore, considering both spatial coverage and retrieval accuracy, the POSP AOD demonstrates a significantly better response capability to pollution events in NCP compared to MODIS DB.

The northern part of India is the Indo-Gangetic Plain, which is a major agricultural area and is densely populated, while the southern part is the Deccan Plateau, which is a large, elevated area known for its unique geology and semi-arid climate. India as one of the most polluted regions globally, has long been a focus of aerosol research [68]. The unique geography of North India, bordered by the Himalayas, leads to persistently high pollution levels in this area [65], [69]. In addition, the northern India close to the Thar Desert is prone to experience dust transport, and the north provinces such as Punjab and Haryana may experience large-scale crop straw burning after the autumn harvest, producing substantial organic carbon aerosols [70].

Fig. 6(a) shows that on March 12, 2022 the pollution was mainly concentrated over central India, while Fig. 6(b) illustrates that on March 17, 2022, high AOD occurred over the Indo-Gangetic Plain in northern India. In Fig. 6(b), an aerosol plume can be distinctly observed originating from eastern

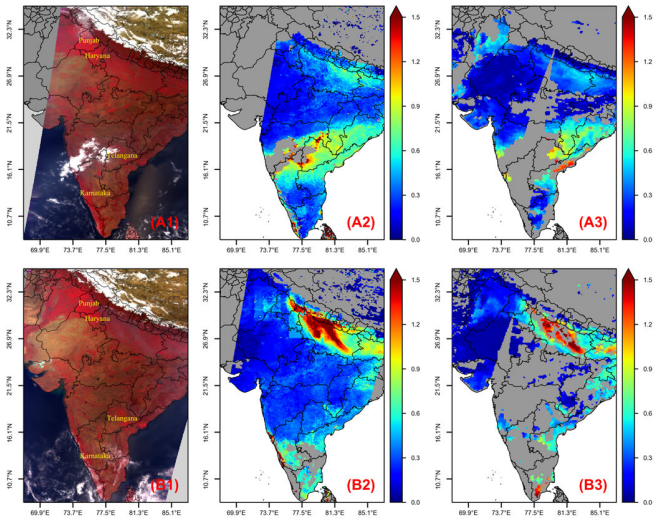


Fig. 6. Same as Fig. 5, but over India on (a1)–(a3) March 12, 2022. (b1)–(b3) March 17, 2022.

Punjab and northern Haryana. The AOD on the west side of the plume is relatively low, while the AOD on the eastern side is significantly higher than in other regions, suggesting that this pollution event is likely originated from eastern Punjab and northern Haryana. The AOD maps in Fig. 6 show that POSP provides much better coverage in North India than MODIS DB and captures the full extent of the aerosol plume on March 12, 2022 and March 17, 2022.

A comparison of AOD retrieved from POSP and MODIS DB with AERONET AOD for major AERONET sites in northern India and adjacent areas is provided in Table S10. Notably, Amity_Univ_Gurgaon and IIT_Delhi are positioned closely on the western Gangetic Plain, while Pokhara is situated further north in central Nepal. Except for the March 17, 2022 data at Amity_Univ_Gurgaon, POSP AOD are generally higher but more closely with AERONET AOD than MODIS DB AOD.

V. Discussion

A. Information Content Analysis

To assess the effect of adding polarimetric measurements on aerosol retrieval, we used the unified linearized vector radiative transfer model (UNL-VRTM) to simulate the synthetic measurements [71]. The UNL-VRTM is primarily based on the VLIDORT radiative transfer model [72] and includes the linearized Mie and linearized T-matrix codes for calculating aerosol scattering [73], as well as gas absorption calculation. By comparing the difference in information content before and after adding polarimetric measurements, we can quantitatively evaluate the effect of adding polarimetric measurements on the retrieval process.

Previous studies have demonstrated that the influence of ALH on apparent reflectance increases as the wavelength decreases [56]. To reduce the impact of ALH on the retrieval results, the 380 and 410 nm bands, which are significantly affected by ALH, are not included. The aerosol models used in the simulations are listed in Table S6, and the surface parameter is provided in Table S7. Two different observation geometry scenarios in Table S8 were considered, which

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represent typical observations. The detailed description of the synthetic parameters is provided in the supplementary document.

Fig. S2 presents the scattering phase functions for different aerosol models. Similar to the results from [63], changes in the polarization phase function mainly occur when scattering angles between 30° and 150°. The intensity scattering phase function (P_{11}) exhibits similar variations across the VIS to NIR with significant differences only appearing in the SWIR when scattering angles exceed 150°. In contrast, the polarization scattering phase function (P_{12}) shows considerable differences across various wavelengths. This suggests that adding polarimetric measurements can enhance the information content for the coarse-dominated aerosol retrieval.

Two typical surface types are considered in the simulations: grassland and bare soil, where surface reflectance spectra from the ASTER spectral library are used. The results are presented in Fig. S3. The main reason for using only these two surface types is that the primary goal of this experiment is to investigate the ability of POSP to retrieve AOD, so it is unnecessary to consider all surface types.

When calculating the DFS, prior errors for each parameter must be considered. In this study, prior errors provided by [35] are used. For retrieval parameters, the prior errors for $V_0^{(f)}$ and $V_0^{(c)}$ are set to 80%, while the prior error for the BRDF kernel coefficients is 80% [74]. The prior error for the BPDF parameters is set to 40% [53]. For nonretrieval parameters, the prior errors are as follows: for $r_{\text{eff}}^{(f)}$, $v_{\text{eff}}^{(f)}$, $n_r^{(f)}$, and $n_i^{(f)}$ they are 15%, 15%, 0.025%, and 50%, respectively; for $r_{\text{eff}}^{(c)}$, $v_{\text{eff}}^{(c)}$, $n_r^{(c)}$, and $n_i^{(c)}$, they are 35%, 35%, 0.04%, and 50%, respectively. The observation errors for POSP are provided by [38], and it is assumed that there is no correlation between errors across different bands The definitions of all the aforementioned parameters can be found in the supplementary documentary, and the rationale behind solving for the DFS of V_0 instead of the DFS of AOD at each wavelength can be found in [35].

Fig. 7 presents the DFS for each retrieval parameter under conditions that consider surface directional reflection and utilize both intensity and polarimetric measurements (Li et al. [35] pointed out if DFS $>$ 0.5, the retrieval of parameter could be carried out; if DFS $>$ 0.75, the retrieval of parameter could be regarded as well retrieved). The results indicate that aerosol content is more easily retrieved over vegetation than over bare soil (higher DFS values), which is in line with previous conclusions [23]. Additionally, a comparison between the DFS obtained using both intensity and polarization and those using only intensity shows that the improvement in information on fine-dominated aerosol is significantly lower than that on coarse-dominated aerosol. In Fig. 7, The left column represents results using both intensity and polarization, while the right column represents results using only polarization. We compared only the aerosol-related V_0, where the error bars indicate the possible range of DFS under different observation and AOD conditions, and the height of the column represents the average DFS. Compared to the right column, the DFS range and average value in the left column are both higher. Therefore, we can conclude that adding polarization

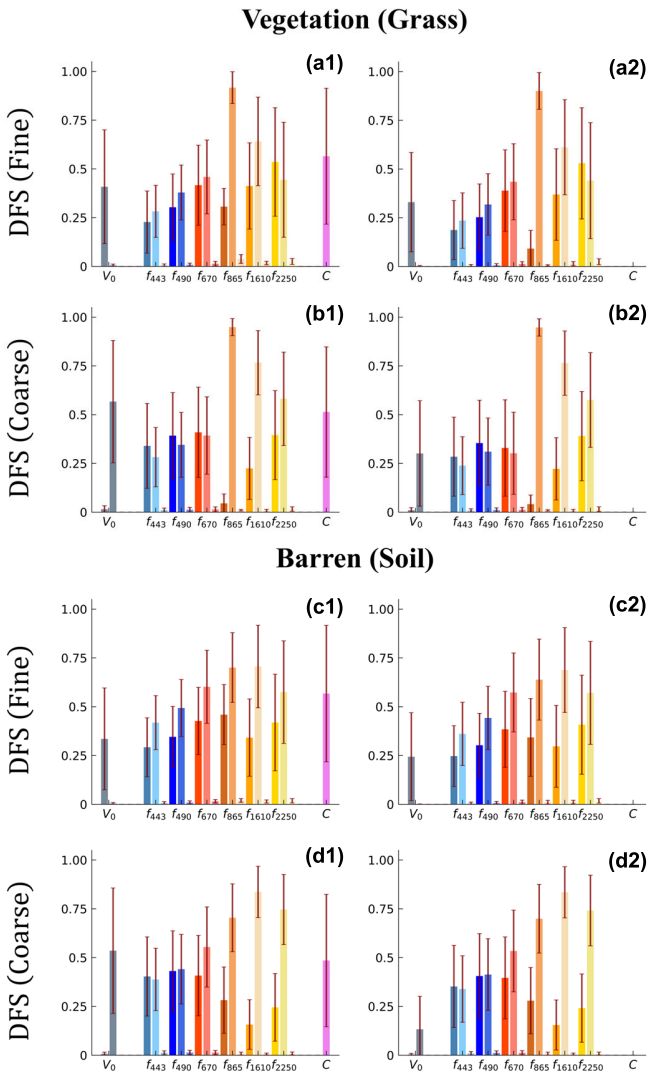


Fig. 7. DFS for each retrieved parameter is analyzed considering surface directional reflectance for two scenarios. (1) Using both intensity and polarimetric measurements. (2) Using only intensity measurements. (a) and (c) Correspond to vegetation (grass), while (b) and (d) correspond to bare soil. The upper panels represent the fine-dominated case, and the bottom panels represent the coarse-dominated case. The error bars indicate the range of possible DFS values for varying observational geometries and AOD conditions (varying observational geometries and AOD conditions can be found in Supplementary).

observations increases the information content for aerosol retrieval.

Fig. 8 presents the relations between the information content of the retrieved parameters and the aerosol loading. To visually compare the information content of BRDF parameters for different aerosol loading, we use the equivalent surface parameter S, as defined by [35] in the form of

$$\text{DFS}(S) = \frac{1}{3n+1}\big(\text{DFS}\big(f_{\text{iso}}(\lambda_1)\big) + \text{DFS}\big(f_{\text{vol}}(\lambda_1)\big) + \text{DFS}\big(f_{\text{geo}}(\lambda_1)\big) + \cdots + \text{DFS}(C)\big) \tag{11}$$

where n represents the number of bands involved in the retrieval. Following [35], we assume that the information content of BRDF parameters is equal to the DFS of S.

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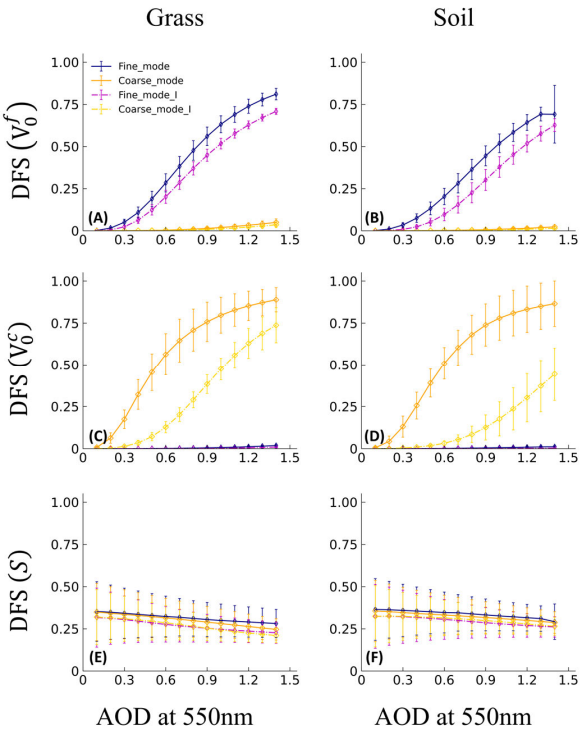


Fig. 8. DFS for the retrieval of aerosol content and surface parameter S for AOD varying from 0.1 to 1.4 at 550 nm. (a), (c), and (e) Corresponds to vegetation (grass) and (b), (d), and (f) to bare soil. The error lines represent the standard deviations corresponding to the results obtained for different observation geometries.

As mentioned above, the contribution of aerosol and surface reflectance to the apparent TOA reflectance is influenced by the AOD. Here, we use the DFS of aerosol volume concentration instead of the corresponding DFS of AOD, where $V_0^{(f)}$ represents fine-mode dominated aerosols, and $V_0^{(c)}$ represents coarse-mode dominated aerosols. The higher the AOD, the greater the contribution of aerosols, which result in an increase in the DFS. Therefore, in Fig. 8, as the AOD increases, the DFS of aerosol loading increases accordingly [Fig. 7(a)–(d)], while the DFS of surface parameters decreases [Fig. 7(e) and (f)]. In addition, Fig. 8 shows that after adding polarimetric measurements, the DFS for coarse-mode dominated aerosols [Fig. 8(c) and (d)] increases more with AOD than for fine-mode dominated aerosols [Fig. 8(a) and (b)]. When only intensity observations are used, the slopes of the DFS corresponding to fine-mode dominated and coarse-mode dominated aerosols with respect to AOD are similar. Additionally, by comparing Fig. 8(c) and (d), we find that for retrieving coarse-mode dominated aerosols, the increase in DFS due to adding polarization over surfaces dominated by grass is greater compared to those dominated by soil. Moreover, after adding polarization, the DFS over grass and soil become roughly equivalent, which confirms that when AOD is high, the effect of surface type for AOD retrieval can be ignored.

B. Error Characteristic Dependence

To investigate the reason for the bias, we compared the retrieval results from the POSP algorithm proposed in this study with those obtained using only intensity measurements [37], for the same observation period (January 1–June 30, 2022). The box plot in Fig. 9 illustrates the impact of adding polarization for various atmospheric and surface conditions. The first quartile ($Q1$) and third quartile ($Q3$) values together estimate the uncertainty range of the error, while the median shows the trend of retrieval error variations for different conditions which can reveal the reason and extent of retrieval bias. The interquartile range (IQR), or the range between $Q1$ and $Q3$, displays the middle 50% of the bias distribution, which represents the "typical" fluctuation of bias.

Fig. 9(a) shows how AOD bias changes with AERONET AOD and statistical metrics are provided in Table I. Compared to the retrievals using only intensity, the addition of polarization results in a reduction of the uncertainties, especially for large AOD (represented by the boxplot's upper and lower limits).

In Fig. 9(b), the effect of adding polarization on the bias is examined for different vegetation conditions as expressed by NDVI. Using the median of the boxplot as a bias indicator, it is apparent that intensity-only retrievals underestimate AOD for high NDVI, which is substantially improved by adding polarization. Additionally, in low NDVI conditions (i.e., barren), the addition of polarization reduces the uncertainty of the retrieval results. The impact of adding polarization under low NDVI conditions is similarly reflected in subsequent comparisons. The comparison in Fig. 10 shows that the retrieval accuracy improves significantly after adding polarization, with the RMSE decreasing from 0.106 to 0.083.

In Fig. 9(c), the variation of the POSP AOD bias with scattering angle is presented. The intensity-only retrievals deviate more from the zero-bias line when scattering angle less than 100° or greater than 140° than the retrievals with both intensity and polarization. Thus, polarimetric measurements effectively reduce the impact of scattering angles, thereby lowering uncertainties arising from varying observation conditions.

In Fig. 9(d), the effect of aerosol particle size, represented by the AE, on the bias distribution of POSP AOD is presented. For larger particles (AE $<$ 0.5), adding polarimetric measurements effectively reduces the lower limits of the boxplot, indicating that the underestimation using only intensity measurements has been mitigated. This is consistent with the information content analysis results in Section V-A. Therefore, we can conclude that adding polarization measurements decrease the uncertainty of coarse-dominated aerosols retrieval. Meanwhile, adding polarization makes the AOD bias closer to 0 under various aerosol and surface conditions and addresses the issue of large retrieval errors in certain scenarios when relying solely on intensity measurements.

C. Improvement of Adding Polarization

The comparison with the POSP AOD retrieved using only intensity revealed an underestimation over bare soil [55]. AOD retrieval over bright surfaces has always been one of the challenges in aerosol retrieval [23]. The addition of polarimetric measurements leads to a substantial improvement as shown in Fig. 10. After adding polarization, the R^2 increased from 0.57 to 0.73, and the fraction within EE and the fraction

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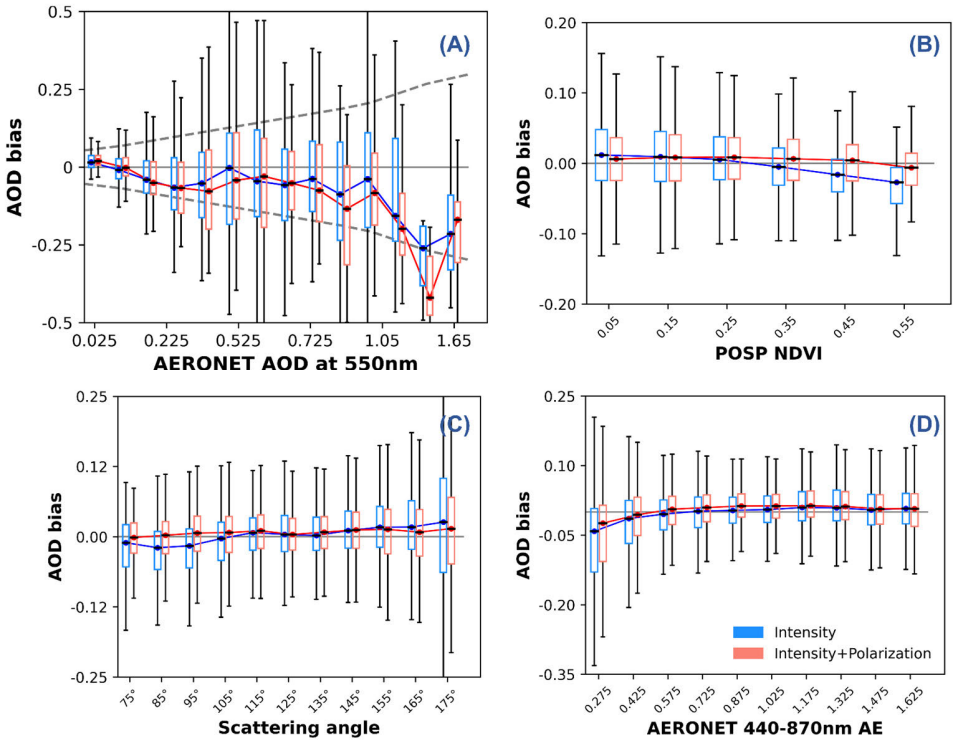


Fig. 9. AOD bias (i.e., POSP AOD-AERONET AOD) variation with (a) AERONET AOD at 550 nm, (b) POSP NDVI, (c) scattering angle, and (d) AERONET 440–870 nm AE. The black dots and error bars represent the median (dots), maximum and minimum values of the 95% confidence of the AOD bias (error bars). The top and bottom edges of the box represent the first quartile ($Q1$) and third quartile ($Q3$), respectively, and the box length represents the 50% of bias range, known as the IQR. The whiskers above and below the box indicate the range of error distribution, with the upper whisker typically extending to the value of Q3 + 1.5 × IQR and the lower whisker to the value of $Q1$ − 1.5 × IQR. The retrieval results using both intensity and polarization are presented in red, and those using only intensity are presented in blue.

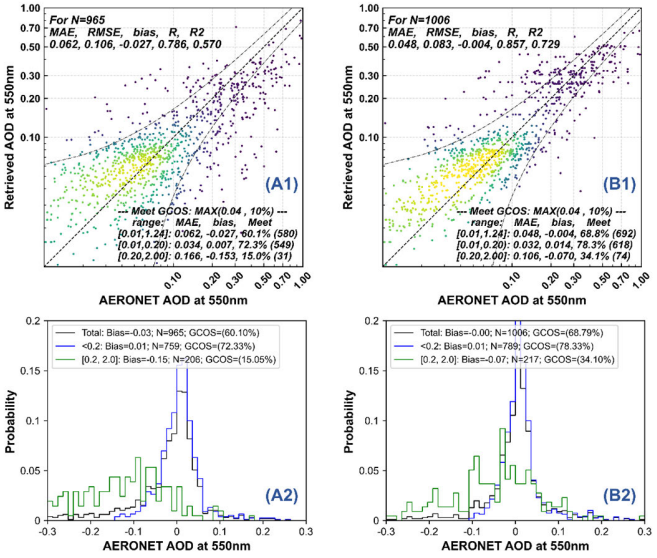


Fig. 10. Same as Fig. 2, but only for AOD retrieved over bare soil using (a1) and (a2) previous POSP AOD algorithm using only intensity measurements and (b1) and (b2) updated POSP AOD retrieval using both intensity and polarization.

within GOCS both improved by nearly 10%. When comparing retrievals for different AOD ranges, we found that for 0.2 < AOD < 0.7 the fraction within GOCS doubled (from 15.54% to 36.45%). We can also observe that for 0.1 < AOD < 0.5, the addition of polarization brings more points below the EE line closer to the 1:1 line, which is reflected in improved accuracy.

From the comparison of AOD bias and AE in the previous section, we already know that the enhancement in accuracy due to polarization is primarily concentrated on coarse-mode dominated aerosols. Over bare soil, including deserts, the aerosol is often dominated by coarse-dominated aerosols, so the improvement in retrieval accuracy for bare soil should also be attributed to the enhanced accuracy in retrieving coarse-dominated aerosols.

VI. Conclusion

Previously, as one of the few single-angle polarimeters currently in orbit, retrieval algorithms based on POSP only utilized intensity measurements without considering polarimetric measurements. We successfully developed an aerosol algorithm by integrating both intensity and polarization from POSP within our optimization framework. With two blue bands and unique SWIR polarization capabilities, the potential of POSP for aerosol detection has been successfully explored. The improvement in retrieval accuracy brought by POSP's polarimetric measurements was systematically examined, for the first time, offering valuable insights for the development of algorithms for other single-viewing polarimeters. Using the Maignan BPDF model, POSP AOD data from January to June 2022 were processed.

Validation showed good agreement with AERONET products, for AOD with an R of 0.899, an RMSE of 0.086, a bias

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of 0.003, fraction within EE of 77.64%, and fraction within GCOS of 58.73%; for AE with an RMSE of 0.404, an MAE of 0.303, a bias of −0.068, and fraction within the EE of 71.61% when $AOD_{550} > 0.4$; for SSA with an RMSE of 0.404, an MAE of 0.044, a bias of −0.02, fraction within the EE of 76.06% and fraction within the GCOS of 54.93% when $AOD_{550} > 0.4$. Additionally, global AOD results derived from POSP were compared with MODIS AOD, indicating good agreement in spatial variation between the two datasets.

Further validation was conducted by comparing POSP AOD with MODIS DB AOD over highly polluted regions (northeastern China and northern India). The results showed that POSP AOD is in better agreement with AERONET AOD than MODIS DB AOD and provided better spatial coverage.

By comparing the information content (DFS) obtained through the optimal estimation theory, we found that adding polarimetric measurements effectively enhances the information content of aerosol parameters, especially for coarse-dominated aerosols. To investigate this further, we compared the retrievals over bare soil dominated by coarse aerosols before and after adding polarimetric measurements. The results demonstrate that adding polarimetric measurements effectively improves retrieval accuracy. Although the AOD was successfully retrieved from POSP data, and the results are of good quality, as indicated by the validation metrics mentioned above, there remains substantial room for improvement. Additionally, considering the limitations of single-angle observations, we plan to apply our optimization framework to the multiangle polarimeter DPC, which flies on the same platform as POSP.

Finally, there is currently no cloud detection algorithm specifically developed for POSP. Enhancing cloud detection accuracy will be crucial for obtaining more reliable results. Overall, this study pioneered the development of a retrieval algorithm that integrates both intensity and polarimetric measurements from POSP, revealing the advantages of polarization over intensity-only, resulting in outcomes with higher accuracy.

Supplementary Data

Figs. S1–S5 and Tables S1–S10.

Acknowledgment

The AErosol RObotic NETwork (AERONET) data used for this study is available at https://aeronet.gsfc.nasa.gov/ (AERONET website). They would like to thank staff at all sites for the data used for this study. They also thank the Level 1 data of particulate observing scanning polarimeter (POSP) provided by China Centre for Resources Satellite Data and Application (https://data.cresda.cn/). ERA5 data were generated using Copernicus Climate Change Service Information (https://cds.climate.copernicus.eu/).

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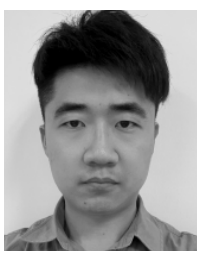
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